

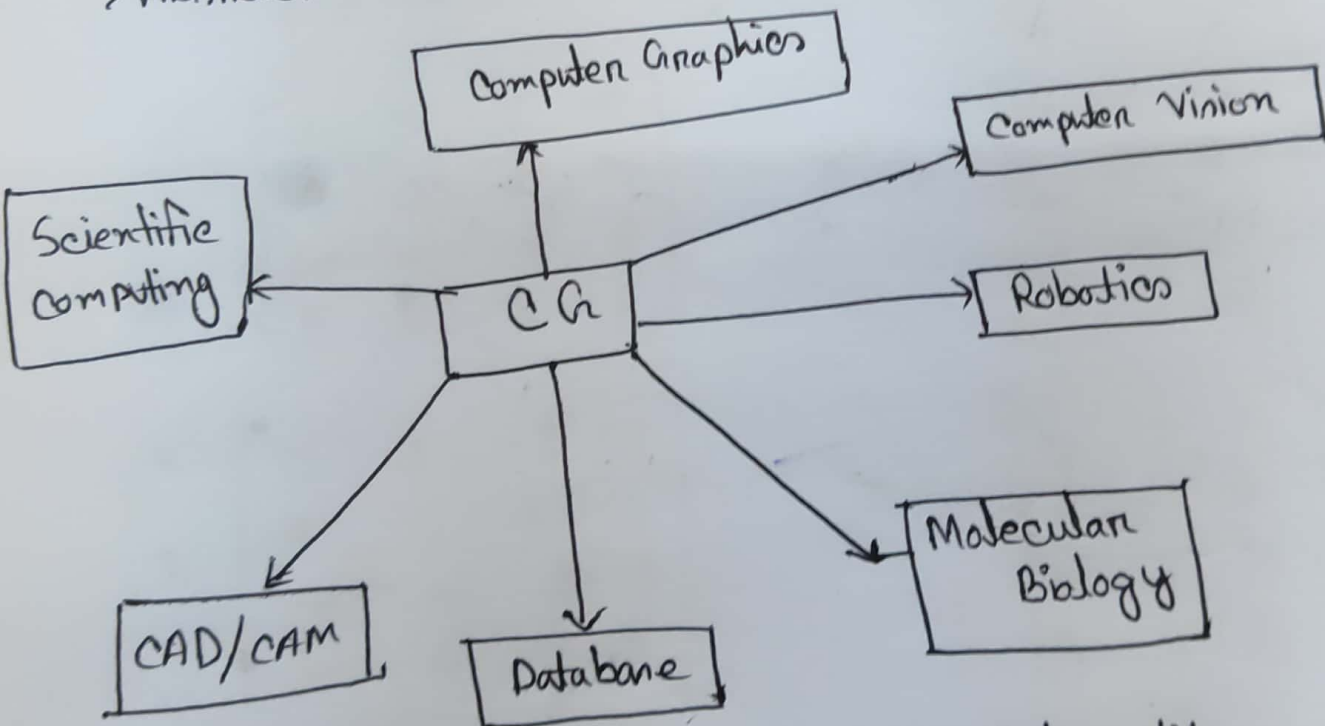
Segment-8

Segment-7

Computational Geometry

- Study of algorithms for geometric problems.
- Deals with discrete shapes: points, lines, polyhedra, polygons, meshes.

→ Abstraction of problems in different applied areas:



Application:

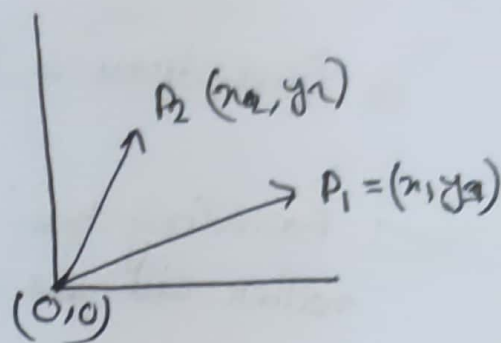
- Occlusion, visibility, augmented reality
- Collision detection, motion or assembly planning
- drug design, database, GIS, layout, fluid dynamics, etc.

• Cross Product

$$u \times v = uv \sin \theta$$

$$P_1 \times P_2 = x_1 y_2 - x_2 y_1$$

$$= -P_2 \times P_1$$



① Given directed line segments $\overrightarrow{P_0 P_1}$ and $\overrightarrow{P_0 P_2}$ determine whether $\overrightarrow{P_0 P_1}$ is clockwise from $\overrightarrow{P_0 P_2}$ with respect to point P_0 ?

→ Compute, $\pi = P_1 \times P_2$

if $\pi > 0$; $\overrightarrow{P_0 P_1}$ is clockwise from $\overrightarrow{P_0 P_2}$.

if $\pi < 0$; $\overrightarrow{P_0 P_1}$ is counterclockwise from $\overrightarrow{P_0 P_2}$.

if $\pi > 0$; $\overrightarrow{P_0 P_2}$ is counterclockwise from $\overrightarrow{P_0 P_1}$.

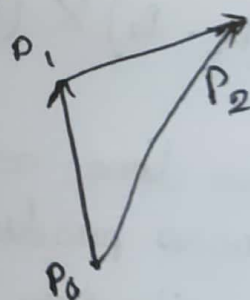
if $\pi < 0$; $\overrightarrow{P_0 P_2}$ is clockwise from $\overrightarrow{P_0 P_1}$.

② Given two line segments $\overrightarrow{P_0 P_1}$ and $\overrightarrow{P_1 P_2}$. if we traverse $P_0 P_1$ and then $P_1 P_2$, do we make a left turn at point P_1 ?

→ compute $\pi = (P_2 - P_0) \times (P_1 - P_0)$

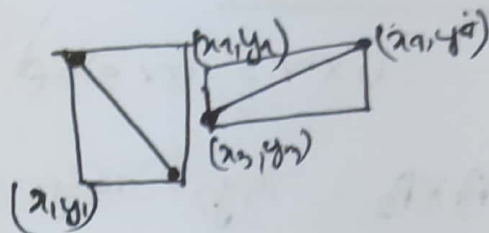
if $\pi > 0$, then we make a right turn at P_1 .

if $\pi < 0$, then we make a left turn at P_1 .



③ If two segments Intersect?

→ first draw a bound box



Case 1: Bounding boxes do not intersect, neither will the segment.

Bounding boxes do not intersect if and only if,

$$x_4 < x_1 \vee x_3 > x_2 \vee y_4 < y_1 \vee y_3 > y_2$$

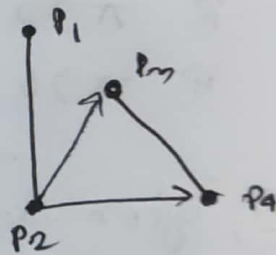
Case 2: Bounding boxes intersect; the segment may or may not intersect.

need further check:

① Line segments do not intersect if and only if one segment lies entirely to one side of the line containing the other segment.

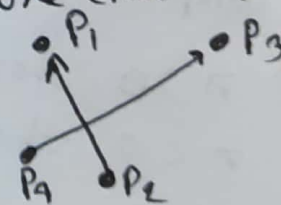
$$(P_3 - P_2) \times (P_1 - P_2) \text{ and}$$

$(P_4 - P_2) \times (P_1 - P_2)$ are both positive or negative

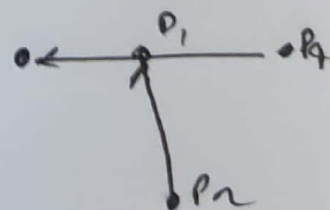


② Two line segments intersect iff each of the two pairs of cross products have different signs (or one cross product is 0).

$$(P_1 - P_4) \times (P_3 - P_4) \text{ and } (P_2 - P_4) \times (P_3 - P_4) \rightarrow$$



$$(P_3 - P_2) \times (P_1 - P_2) \text{ and } (P_4 - P_2) \times (P_1 - P_2) \rightarrow$$



• Convex polygon:

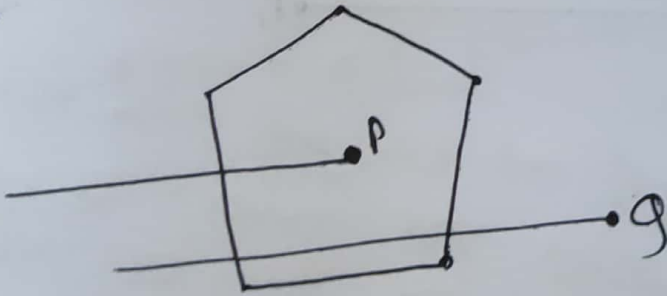
- Convex polygon has every angle less than 180° and a line between any two points are in the polygon.

- Non convex polygon ~~has every~~ may have one or more in degrees greater than 180° .

To determine whether a given point lies outside or inside a given polygon?

⇒ Draw a line along the x-axis from the given point in any direction.
→ If the line intersects only once with any edge of the polygon then the point lies inside the polygon else it lies outside the polygon.

Example:

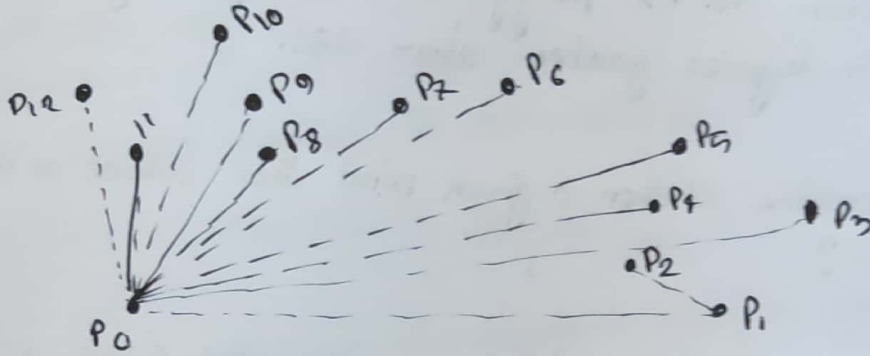


Convex Hull - Problem:

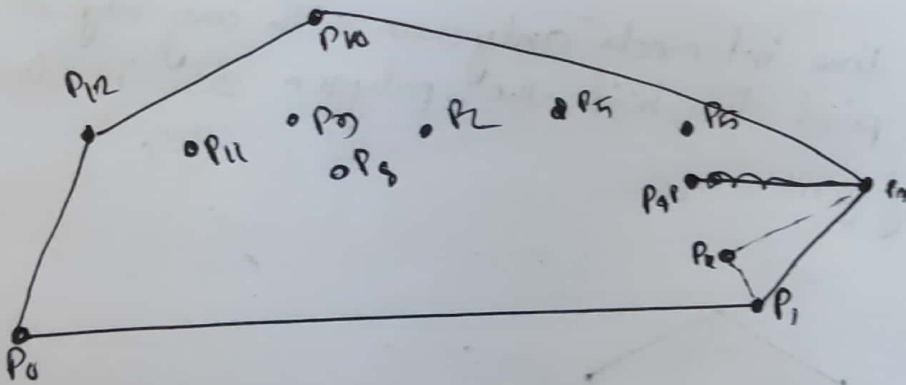
- Given n points on plane $P_1, P_2, \dots, P_n$, find the smallest convex polygon that contains all points $P_1, P_2, \dots, P_n$.

This problem is solved by Graham Scan:

Example:



Ans:



First $P_0 P_1$ and $P_1 P_2$ will be drawn. তারপর প্রতি
বিন্দু থেকে $P_0 P_1$ রেখার দিকে ঘুরে 180 ডিগ্রি
এর কোণ হতে কিনা। হলে এই বিন্দু
খارجে বেরিয়ে আসবে অন্যথায়
এই point polygon তৈরি থাকবে।

Complexity:

Time Complexity: $O(n \log n)$

Number Theory:

⇒ Numbers used in Cryptography:

- Non-negative integers

Prime: A positive integer greater than 1 is prime if and only if it is evenly divisible by only 1 & itself.

Divisor: If a & b are positive integers and $b \neq 0$, b is a divisor of a (b divides a) if $a = mb$ for some integer m , such that $a/b = m$.

• Properties of Divisors:

Notation (1): $b|a$ means b divides a with no remainder or b is a divisor of a .

⇒ if $a|k$ and $a|l$, then $a|(mk + nl)$ for arbitrary m & n

Example: If $a = 6$; $k = 36$; $l = 54$; $m = 2$ $n = 4$

So, $6|36 = 6$ and $6|54 = 9$ so does $6|(2 \times 36 + 4 \times 54)$

• Divisor THEOREM:

Theorem: Let a, b and c be the integers. Then:

① $a|b \wedge a|c \rightarrow a|(b+c)$

② $a|b \rightarrow a|bc$

~~③ $a|b \wedge a|c \rightarrow a|b-c$~~

③ $a|b \wedge b|c \rightarrow a|c$



Example:

$$\textcircled{1} 17|34 \wedge 17|170 \rightarrow 17|204$$

$$\textcircled{2} 17|34 \rightarrow 17|340$$

$$\textcircled{3} 6|12 \wedge 12|144 \rightarrow 6|144$$

- Greatest Common Divisor:

→ The greatest common divisor (gcd) of two natural numbers is the largest divisor shared by these two numbers.

- Modular Arithmetic:

Several important cryptosystems make use of modular arithmetic. This is when the answer to a calculation is always in the range $0-m$ where m is the modulus.

Example:

$$17 \bmod 11 = 6$$

$$-1 \bmod 11 = 10$$

$$-11 \bmod 11 = 0$$

$$25 \bmod 5 = 0$$

⇒ Two numbers ' r ' and ' s ' said to be "congruent mod m " if $r \bmod m = s \bmod m$.

Example: if $6 \bmod 5 = 16 \bmod 5$

then, $6 \equiv 16 \bmod 5$

- The differences between r and s will be a multiple of m
So $r - s = km$ for some value of k
eg. $16 - 10 = 2 \times 5$

- Computational complexity of calculating mod is $O(b^2)$
- Computational complexity of performing multiplication mod m is $O(b^2)$
- Computational complexity of calculating $x^n \bmod m$ is $O(b^3)$ where b is the size of n .

→ Modular Division:

$$a \div b \bmod c$$

→ need to multiply 'a' by inverse of $(b \bmod c)$.

$$\text{For inverse of } (b \bmod c); (b * k) \bmod c = 1$$

Example:

$$\Rightarrow 5 \div 3 \bmod 11?$$

$$\text{inverse of } 3 \bmod 11, (3 * 4) \bmod 11 = 1$$

$$\text{Thus, } 5 \div 3 \bmod 11$$

$$= 5 * 4 \bmod 11$$

$$= 9 \bmod 11$$

It's relatively easy to find the inverse of $x \bmod m$ using Euclid's algorithm.

It's computational complexity $O(b^3)$; where b is the size of m .

• 'x' doesn't have an (inverse mod m) unless 'x' and 'm' are co-prime.



- Euclid's GCD Algorithm:

⇒ Applied formula: $\text{gcd}(a, b) = \text{gcd}(b, a \bmod b)$

Example:

$$\text{gcd}(412, 260) = 4$$

a	412	260	152	108	44	20	4
b	260	152	108	44	20	4	0

⇒ The overall running time of Euclid GCD is proportional to the number of recursive call it make.

☐ A number $n \geq 2$ which isn't prime is called composite.

Primality Testing:

Test if 139 and 143 are prime

Relatively Prime numbers:

Two numbers are relatively prime if their gcd = 1

Factor: a number that is multiplied by another to give product

$$7 \times 8 = 56$$

↓
Factors

a number that divides evenly into another.

$$56 \div 8 = 7$$

↓
Factor

Hundreds Chart:

1	2	3	4	5	6	7	8	9	10	① even-1
11	12	13	14	15	16	17	18	19	20	② even; multiple of 2
21	22	23	24	25	26	27	28	29	30	③ even - multiple of 3
31	32	33	34	35	36	37	38	39	40	④ multiple of 5
41	42	43	44	45	46	47	48	49	50	⑤ multiple of 7
51	52	53	54	55	56	57	58	59	60	⑥ multiple of 11
61	62	63	64	65	66	67	68	69	70	
71	72	73	74	75	76	77	78	79	80	
81	82	83	84	85	86	87	88	89	90	
91	92	93	94	95	96	97	98	99	100	

Primes number from 1 to 100 are:

2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97.

